



give me viva que and ans on virtualisation, vm, hypervisor, and other cloud computing fundamentals

also include docker, kubernetes

Here is a comprehensive set of viva and interview questions with precise answers covering virtualization, virtual machines (VMs), hypervisors, foundational cloud computing concepts, Docker, and Kubernetes.^{[1] [2] [3]}

Virtualization, VMs, and Hypervisors

Q1: What is virtualization in cloud computing?

Virtualization is the process of creating a software-based (virtual) representation of a physical resource, such as a server, storage device, or network. It enables multiple virtual instances to run concurrently on a single physical machine, sharing its underlying resources efficiently. This forms the foundational layer of cloud computing, allowing for dynamic scaling, flexibility, and optimized hardware utilization.^[2]

Q2: What is a Virtual Machine (VM)?

A Virtual Machine is an isolated, software-based environment that acts like a physical computer. It has its own virtual CPU, memory, storage, and network interface, and runs its own complete guest operating system (OS). Because VMs are completely isolated, you can run multiple different operating systems (like Linux and Windows) on the same physical server simultaneously.

Q3: What is a Hypervisor and what are its types?

A hypervisor (or Virtual Machine Monitor) is the software that creates, runs, and manages virtual machines by allocating physical hardware resources to them.^[4]

There are two main types:

- **Type 1 (Bare Metal):** Runs directly on the host's physical hardware without an underlying OS (e.g., VMware ESXi, Microsoft Hyper-V). It provides high performance and is used in enterprise data centers.^{[5] [4]}
- **Type 2 (Hosted):** Runs as a software application on top of a conventional host OS (e.g., VMware Workstation, Oracle VirtualBox). It is generally used for personal computing and testing.

Cloud Computing Fundamentals

Q4: How do you define Cloud Computing?

Cloud computing is the on-demand delivery of IT resources (like computing power, storage, and databases) over the internet. Instead of buying and maintaining physical servers, users rent these resources on a pay-as-you-go basis from cloud providers.^[6]

Q5: What are the key characteristics of Cloud Computing (as per NIST)?

- **On-demand self-service:** Users can provision computing capabilities automatically without human interaction.^[1]
- **Broad network access:** Services are available over the network and accessed through standard mechanisms (phones, tablets, laptops).^[1]
- **Resource pooling:** Provider computing resources are pooled to serve multiple consumers using a multi-tenant model.^[1]
- **Rapid elasticity:** Capabilities can be elastically provisioned and released, scaling outward and inward corresponding with demand.^[1]
- **Measured service:** Cloud systems automatically control, optimize, and report resource usage for billing transparency.^[1]

Q6: What are the primary Cloud Service Models?

- **IaaS (Infrastructure as a Service):** Provides raw virtualized computing resources (servers, storage, networking) over the internet (e.g., AWS EC2).^[7]
- **PaaS (Platform as a Service):** Provides a managed environment to develop, test, and deploy applications without worrying about underlying infrastructure (e.g., Google App Engine).^[7]
- **SaaS (Software as a Service):** Delivers fully functional, end-user applications over the internet (e.g., Gmail, Salesforce).^[7]

Docker and Containerization

Q7: What is Docker and how is it different from a VM?

Docker is an open-source platform used to automate the deployment of applications inside lightweight, portable environments called containers.^[8]

While a VM includes a full guest OS, virtualized hardware, and the application, a Docker container only packages the application and its direct dependencies. Containers share the host system's OS kernel, making them much faster to start, more resource-efficient, and smaller in size compared to VMs.

Q8: What is a Docker Image vs a Docker Container?

- **Docker Image:** A read-only, immutable template containing the application source code, libraries, dependencies, and tools needed to run an application.
- **Docker Container:** A running instance of a Docker Image. When an image is executed, it becomes a container.^[8]

Kubernetes (K8s)

Q9: What is Kubernetes and why is it needed if we have Docker?

Kubernetes is an open-source container orchestration platform. While Docker is excellent for creating and running individual containers, it does not have native tools to manage thousands of containers running across multiple servers. Kubernetes automates the deployment, scaling, networking, load balancing, and health monitoring of these containers.
[3] [8]

Q10: What is a Pod in Kubernetes?

A Pod is the smallest, most basic deployable computing unit in Kubernetes. It encapsulates one or more tightly coupled Docker containers that share the same storage resources, network IP, and configuration on how to run. Kubernetes manages Pods rather than managing containers directly.^[8]

Q11: How does Kubernetes handle automatic scaling and self-healing?

- **Self-healing:** If a container or Pod fails, Kubernetes automatically restarts it. If a node fails, Kubernetes reschedules the Pods onto healthy nodes to ensure the application remains available.
- **Scaling:** Using the Horizontal Pod Autoscaler (HPA), Kubernetes automatically scales the number of Pods up or down based on CPU utilization or other custom metrics.^[3]

Would you like a quiz to test your knowledge of these concepts?

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1. <https://www.scribd.com/document/489465562/Cloud-Computing-Viva-Question-Answer>
2. <https://www.datacamp.com/blog/cloud-computing-interview-questions>
3. https://www.linkedin.com/posts/arungoel0810_interview-devops-kubernetes-activity-7357272576642994177-KBNj
4. <https://in.indeed.com/career-advice/interviewing/hyper-v-interview-questions>
5. <https://attariclassess.in/blog/top-vmware-interview-questions-and-answers>
6. <https://www.geeksforgeeks.org/interview-experiences/cloud-computing-interview-questions/>
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8. <https://www.hirist.tech/blog/top-55-docker-interview-questions-and-answers/>
9. <https://www.sprintzeal.com/blog/cloud-computing-interview-questions>
10. <https://s2-labs.com/blog/cloud-computing-interview-questions/>